

Information Bottleneck-Induced Phase Shift in Symbolic Abstraction Systems

I. Theorem Statement

In any symbolic abstraction system (biological or artificial), there exists a critical latent dimensionality threshold w_c such that:

$$\dim(Z) < w_c \quad \Rightarrow \quad Q(Z) \rightarrow \text{Nonlinear Collapse}$$

Where:

- Z : Internal latent representation
- $\dim(Z)$: Dimensionality of Z
- $Q(Z)$: Symbolic fidelity metric
- w_c : Collapse threshold for symbolic coherence

Second-order collapse behavior:

$$\left. \frac{d^2 Q}{d(\dim Z)^2} \right|_{\dim(Z)=w_c} \gg 0$$

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II. Axioms and Constraints

Information Bottleneck Principle

$$\min I(X; Z) \quad \text{subject to} \quad I(Z; Y) \geq \beta$$

- X : Input symbols
- Z : Latent space
- Y : Target labels or categories
- I : Mutual Information
- β : Fidelity constraint

Symbolic Fidelity Metric $Q(Z)$

Measures coherence in the latent space. Examples:

- Linear probe classification accuracy
 - Mutual information $I(Z; Y)$
 - Topological invariants (Betti numbers, cluster persistence)
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III. Testable Predictions

1. **AI Systems:** Transformers or VAEs will exhibit sharp symbolic degradation when $\dim(Z) \rightarrow w_c$.
 2. **Mutual Information:** $I(Z; Y)$ will inflect sharply at collapse threshold.
 3. **Topological Drift:** Latent space will lose stability (clustering variance increases).
 4. **Cognitive Collapse:** Human symbolic performance degrades nonlinearly under bandwidth limits.
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IV. Empirical Test Design

AI Testing

- **Models:** Transformers, RNNs, VAEs
- **Method:** Gradually compress latent space
- **Metrics:** BERTScore, $Q(Z)$, t-SNE/UMAP drift, $I(Z; Y)$
- **Tools:** MINE, CLUB estimators, probing classifiers

Human Testing

- Symbolic reasoning tasks under memory/attention constraints
 - Collapse criterion: sharp performance drop below cognitive threshold
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V. Simulation Feasibility

- **Frameworks:** PyTorch, TensorFlow
 - **Data:** Symbolic corpora, logic problems, analogical reasoning
 - **Visualization:** MI heatmaps, collapse thresholds, latent topologies
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VI. Theoretical Implications

- Introduces a symbolic phase transition concept in abstraction systems
- Diagnoses collapse before full failure in compressed architectures
- Bridges neural model constraints and cognitive overload behavior
- Supports safety-aware architecture design